# Egg segmentation and tracking: report

**Honesty statement.** No human pixel-level annotation exists. All 'ground truth' masks are SAM-derived pseudo-ground-truth (Task 1 output) and all egg counts, instance verdicts and video identity segments are AI-assisted (Claude Code visual review) and unverified by a human. Metrics are agreement with these references, not accuracy. Claude Code also wrote most of the code; see the AI disclosure at the end. Repository: *not published yet*. Tracking video (Google Drive): **link not created yet** (a footage-free schematic demo video is generated locally by `src/tracking/schematic_video.py`; the original third-party footage is not redistributed).

## 1. Summary

| | dev (88 imgs) | eval (22 imgs) |
|---|---|---|
| S1 (SAM + selector) exact egg count | 0.830 (73/88) | 0.864 (19/22) |
| classical exact egg count | 0.477 (42/88) | 0.545 (12/22) |
| classical instance F1 vs reviewed reference (dev) | 0.416 | not computed (no eval reference) |
| video (T1): detection coverage of expected eggs | 0.963 (dev frames) | 1.000 (held-out frames) |

The pretrained route (S1) is much more reliable than the classical one on this dataset; the classical method is a documented weak baseline that fails whenever the background is as bright as the egg. The two methods are not ranked by pixel accuracy because no independent mask reference exists. S1 under-counts (0.159 of dev images, 0.136 of eval images) with one over-count in 88 dev images, and flags images where nothing passes its gates.

## 2. Data and leakage-aware split

110 images (Roboflow egg export, CC BY 4.0). Exact (SHA) and near-duplicate pairs were linked into groups; groups are split 80:20 with seed 42, giving 88 dev / 22 eval images with no group crossing (`data/split_manifest.csv`). A first split (v1) was contaminated because pilot-inspected images ended in eval; v1 is kept for history and v2 forces those groups to dev. Constraints, grouping decisions and a leakage report are in `data/` and `src/data/splitting.py`; the evaluation images were not inspected or run until both methods were frozen.

## 3. Task 1: pretrained segmentation (S1)

SAM ViT-B (official checkpoint, sha256 recorded in the frozen file) automatic mask generator, 32x32 points, run once for all 110 images; raw candidates are cached (14,400 candidates). A feature selector (area 1-60%, solidity >= 0.95, aspect <= 2, ellipse IoU >= 0.93, predicted IoU >= 0.9, stability >= 0.9, NMS) picks the egg masks; images with no selection are flagged for review, never guessed. A pre-registered set of selector experiments and a root-cause audit are in `experiments/`. An early freeze was withdrawn after display artefacts looked like false positives; the audit found a display defect, not a pipeline bug. A fallback using appearance descriptors and a box-prompted boundary refinement were tested on dev and rejected against pre-registered criteria. Outputs for all 110 images: `outputs/task1_sam_masks/` (15 are empty/flagged: 13 dev, 2 eval).

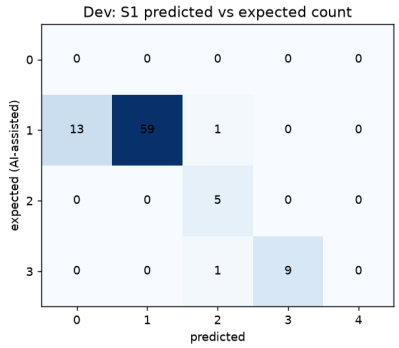
## 4. Task 2: classical segmentation and experiment log

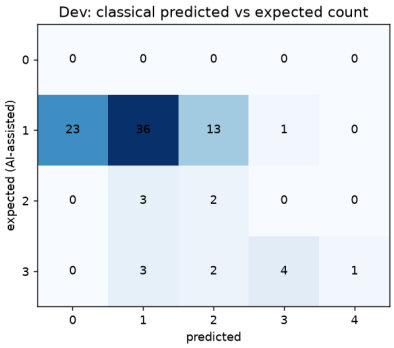
Pre-registered 14 experiments (E00-E13), a greedy keep/discard rule evaluated on dev only, and leave-one-scene-out checks. Final config (frozen as classical_v1, config sha `6696e8cab3ba`): Lab whiteness (L minus chroma), Otsu, 5 px open and close, hole fill, area 1-60%, aspect <= 2.2, border-touching components >= 5% of the image. Every trial, including failures, is in `experiments/classical_log.csv` with full parameters; the table gives the change, why and the keep/discard result (deltas of dev metrics against the comparison base).

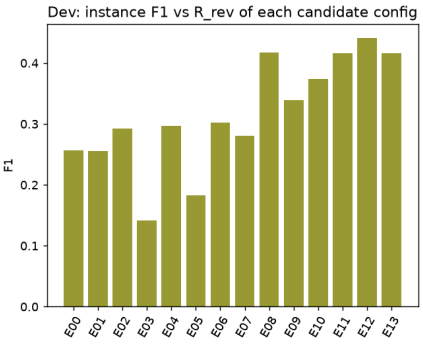
| id | what changed / why | dev deltas vs comparison base (f1 vs R_rev, IoU vs R_rev, exact count) | decision |
|---|---|---|---|
| E00 | baseline: simplest interpretable chain (Lab lightness, Otsu, 5 px open/close, fill holes, area 0.5-60%) | baseline | baseline |
| E01 | HSV value as the foreground map: less affected by chroma than Lab L? | -0.002, +0.000, -0.045 | discarded |
| E02 | whiteness (L - chroma): eggs are bright AND unsaturated; should reject coloured bright regions (cloth, bowls, tile) | +0.036, +0.026, +0.000 | kept |

| id | what changed / why | dev deltas vs comparison base (f1 vs R_rev, IoU vs R_rev, exact count) | decision |
|---|---|---|---|
| E03 | inverted saturation: eggs are nearly unsaturated | -0.115, -0.112, -0.011 | discarded |
| E04 | blur before Otsu to suppress texture (cloth, rubble) that splits regions | +0.005, -0.001, +0.057 | discarded |
| E05 | local threshold for uneven lighting and gradients | -0.110, -0.019, -0.273 | discarded |
| E06 | larger morphology to remove bridges to background and fill gaps | +0.009, +0.012, +0.057 | discarded |
| E07 | smaller morphology to keep thin egg boundaries | -0.012, -0.001, +0.000 | discarded |
| E08 | egg-like shapes are convex: reject ragged regions | +0.125, -0.030, +0.136 | discarded |
| E09 | eggs are compact: reject elongated bars | +0.046, +0.015, +0.045 | kept |
| E10 | reject larger specks and small background blobs | +0.035, +0.003, +0.023 | kept |
| E11 | border-touching components must be large (clipped eggs are large) | +0.043, +0.015, +0.136 | kept |
| E12 | separate touching eggs with a distance-transform watershed | +0.026, +0.009, -0.057 | discarded |
| E13 | shape prior: component must resemble its fitted ellipse | +0.001, -0.136, -0.193 | discarded |

Kept: E02, E09, E10, E11. Near misses decided by the rule: E08 (solidity) lost 0.030 macro IoU (limit 0.010); E12 (watershed) lost 0.057 exact-count rate (limit 0.030). Parameters of every trial: `experiments/classical_log.csv`.

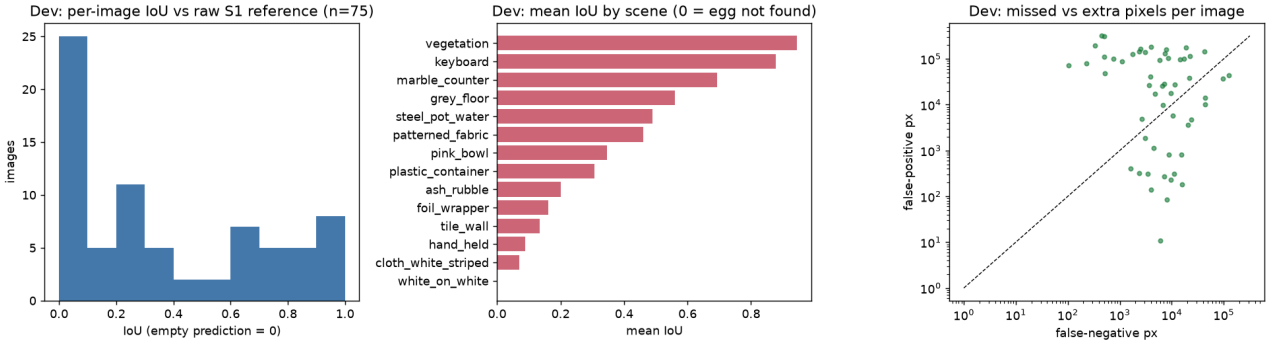


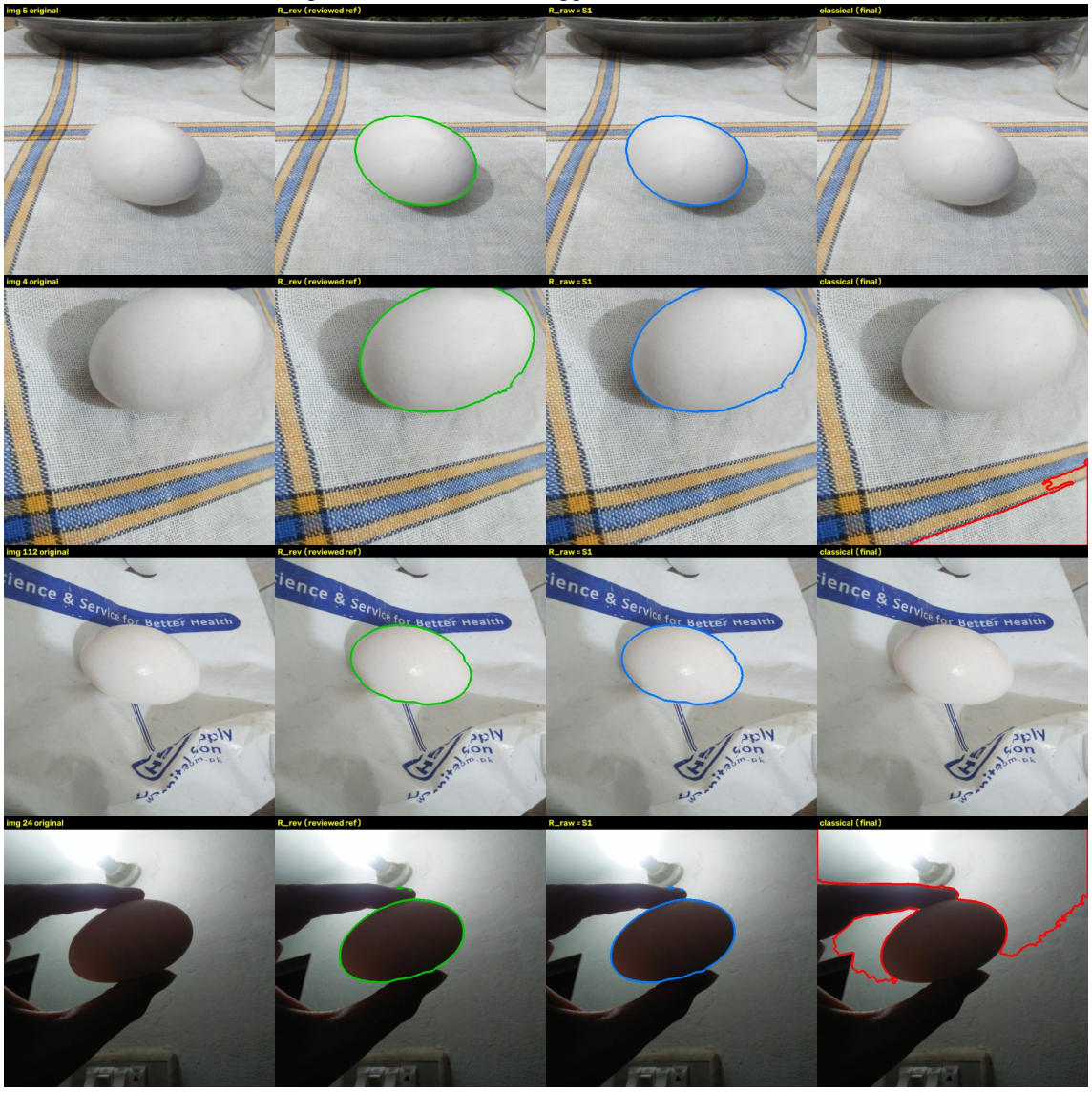



# 5. Task 3: evaluation and analysis (dev; Task 1 masks as reference)

Reference: S1 raw masks (R_raw) and S1 plus 7 AI-assisted additions (R_rev), kept separate. Dev, 88 images, classical vs R_rev: instance P/R/F1 0.442/0.393/0.416, macro IoU 0.363, macro Dice 0.441, count exact 0.477 (over 0.170, under 0.352); vs R_raw macro IoU 0.366. 20 images have the right count with wrong instances (false positive standing in for a miss) against 0 for S1. S1's reference-based numbers are not accuracy (the reference is S1); its independent measure is verdict precision 0.990 / recall 0.876. Per scene the classical F1 is 0 on ash, striped cloth, egg tray, foil, tile and white-on-white, and 1.0 only on a dark keyboard background. Runtime: classical 0.05 s/image CPU; SAM candidates 16.4 s/image GPU plus selector 0.26 s.




Qualitative: worst, best and category overlays (original | reference | S1 | classical). The failure is structural: brightness and low saturation describe the background as much as the egg.




Failure types (details: `experiments/classical_failure_analysis.md`): empty prediction on white-on-white, tile, cloth and pale bowls; backlit egg darker than the wall; touching eggs merged; bright reflective foil or water beats the egg;

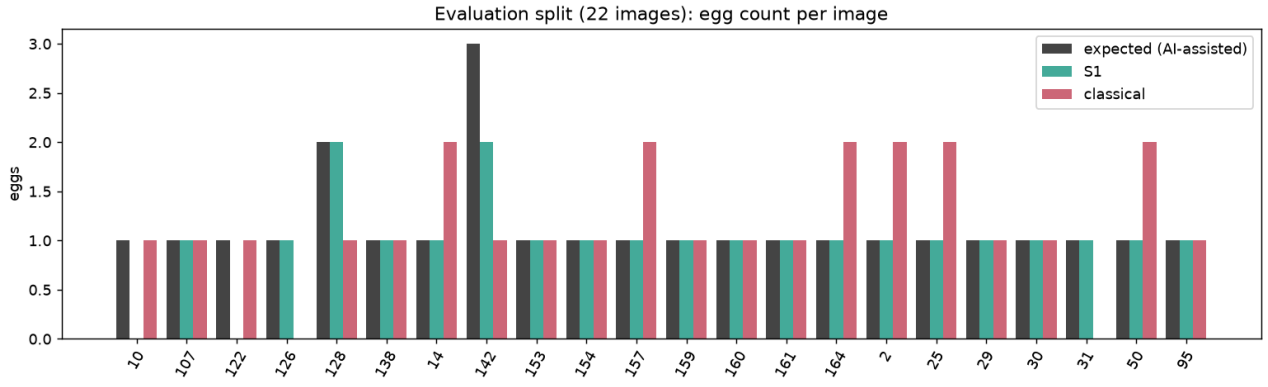
ash, skin and rug regions merged with the egg. Improvement directions (not implemented): learned or colour-texture models, a background model, per-image adaptive features; none fits the classical constraint cheaply.

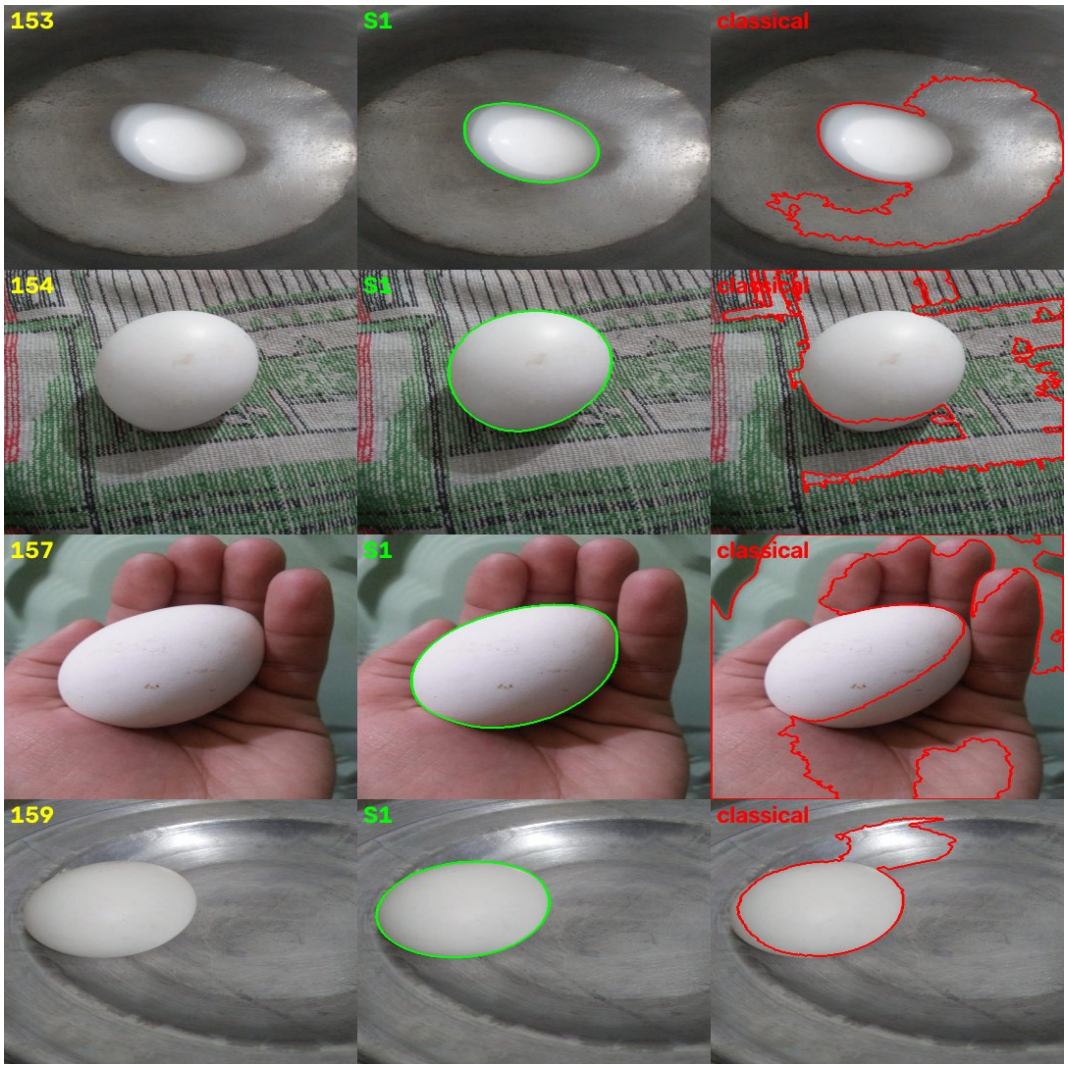
# 6. Frozen methods and final evaluation (22 untouched images, one run)

Both methods were frozen before the evaluation run with hashes, source-file hashes and dependency record (`configs/frozen/S1_v1.json` artifact sha `a6e46871795be95e`, `classical_v1.json` `d8c827fe7d9d12ff`); the run is recorded once in `outputs/eval/RUN_RECORD.json` and cannot be repeated. The protocol was written before freezing (`experiments/final_evaluation_protocol.md`). Eval counts were labelled by AI after the run from raw images but after the predicted counts had been printed (not blind). Note: `classical_log.csv` was appended with the freeze row after the freeze, so its current hash differs from the one recorded in classical_v1.json.

| method | exact | over | under | mean abs err | predicted / expected eggs |
|---|---|---|---|---|---|
| S1 | 0.864 (19/22) | 0.000 | 0.136 | 0.136 | 22 / 25 |
| classical | 0.545 (12/22) | 0.273 | 0.182 | 0.500 | 26 / 25 |

Classical vs S1 raw masks (agreement only, S1 non-empty on 20/22): macro IoU 0.315, Dice 0.439; instances at IoU 0.5: TP 6, FP 20, FN 16. S1 misses: 10 and 122 (flagged), 142 (one of three touching eggs). Classical: empty on 126 and 31; background regions added or merged on most others. Dev and eval are consistent in direction; 22 images allow no significance claim.




153
S1
classical
154
S1
classical
157
S1
classical
159
S1
classical

# 7. Task 4: egg tracking in the video

Video Y4JyV94y2_A (AV1, 1920x1080, 24 fps); experiment interval frames 240-999 (title and explanation cards excluded). The frozen classical config also segments the wall, shirt and hand on this video, so a video-specific detector (grey >= 200, saturation <= 60, morphology, area/solidity/aspect/ellipse-fill filters; fixed a priori) is used, plus a gated Hungarian tracker in two variants: T0 (gate grows with missed steps) and T1 (T0 plus the scene prior 'two eggs'). Sample: every 6th frame (4 Hz, 127 frames), alternate 1-second blocks as dev / held-out (disclosure: the annotator saw all frames; neither is unseen).

| | T0 | T1 |
|---|---|---|
| IDs for 2 eggs | 4 | 3 |
| ID switches | 3 | 2 |
| Q through hand carry 480-558 keeps ID | no | yes |
| Q through hand carry 648-750 | no | no |
| P through pick-up/spin 750-816 | no | no |

Detection (separate from identity): coverage 0.963 on dev frames (4 misses: caption arrow touching an egg at frames 336/342, hand covering an egg at 732/738), 1.000 on held-out, 0 extra detections. Proxy identity evaluation against a single-annotator AI-assisted description (`data/video/identity_segments.csv`): three occlusion events only, so these are illustrations, not rates. **A real failure remains in both variants:** when the hand picks up one egg near the other, the track continues onto the wrong (stationary) egg; two identical eggs without appearance cues cannot be disambiguated by position/motion. Runtime: detector 8.3 ms/frame at 960 px (CPU), tracker 0.1 ms/frame; video decoding dominates. Full details: `experiments/video_tracking_results.md`.

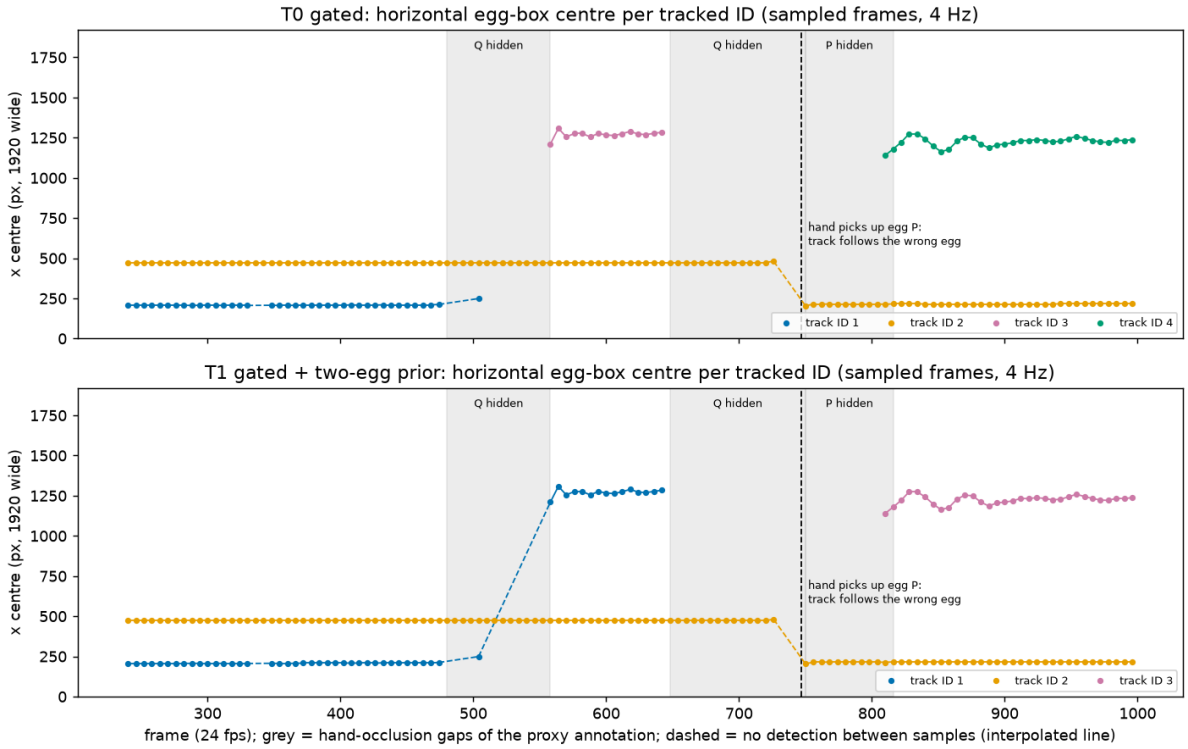



Figure: egg-box centre x per tracked ID from the saved tracks (no third-party footage). The dashed line marks the pick-up at about frame 747 where ID 2 jumps from the picked-up egg to the stationary one (proxy annotation).

Tracking video (Google Drive): **not uploaded; link not available at the time of this report build**. The video to be shared is a footage-free schematic rendering of the tracker output (IDs, frame numbers, trajectory plot, identity-switch annotation); the source footage (MocomiKids, standard YouTube licence) is not redistributed. Replace this line with the verified public link before submission.

# 8. Limitations and open issues

No human-verified labels; references derive from S1; 22 evaluation images; eval counts not blind; video evaluation is a proxy on one clip with 3 occlusion events; the GitHub repository and Drive video do not exist yet; the classical method is a weak baseline; S1 boundaries look smoother/larger than the visible egg edge (unmeasured). Four reviewed additions are first-pass only and seven development images have a bad-boundary S1 candidate excluded from the reviewed

reference (one of them uncertain).

## 9. Reproduce

See README.md (setup, data/weights locations, exact commands). Tests: 197 pass. Evaluation re-run is blocked by design.

## 10. AI-tool disclosure

Claude Code (Anthropic) wrote most code, tests, protocols and analyses under the author's direction and produced the AI-assisted labels by viewing images; these labels are unverified by a human. The author is expected to explain the methods in the interview; withdrawn or rejected steps (premature selector freeze, contaminated split v1, rejected fallback and boundary refinement) are recorded in the logs.